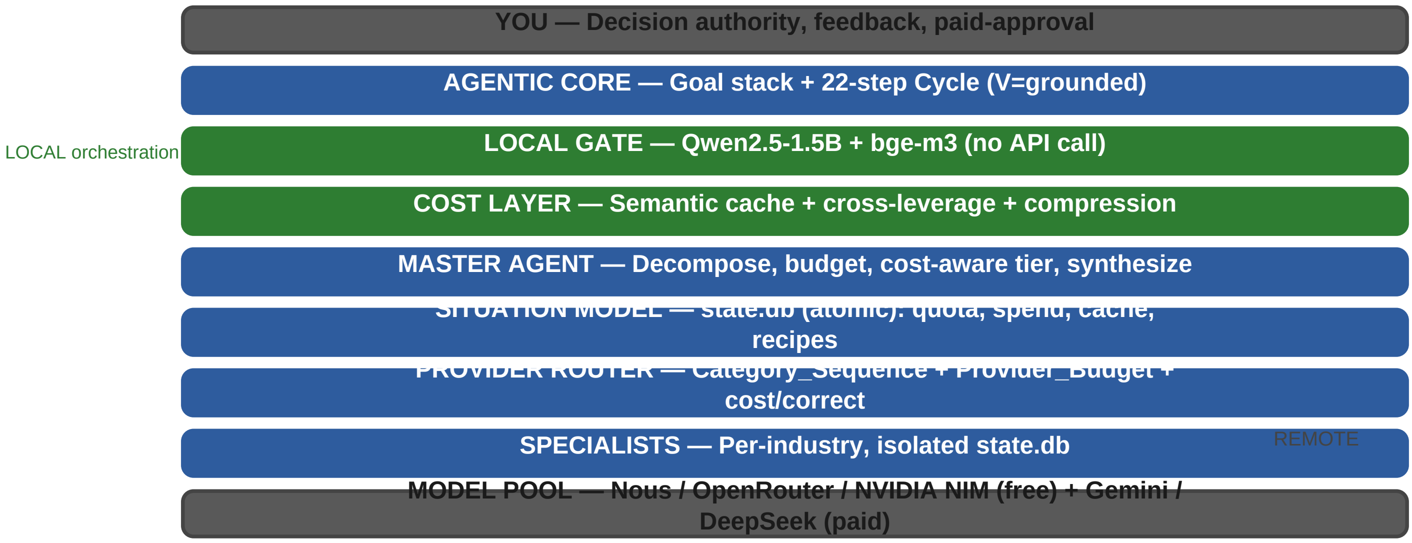
Hermes Meta-Intelligence — Architecture Diagrams

v3.5 · Cost-aware, grounded, self-improving, interruptible agentic system

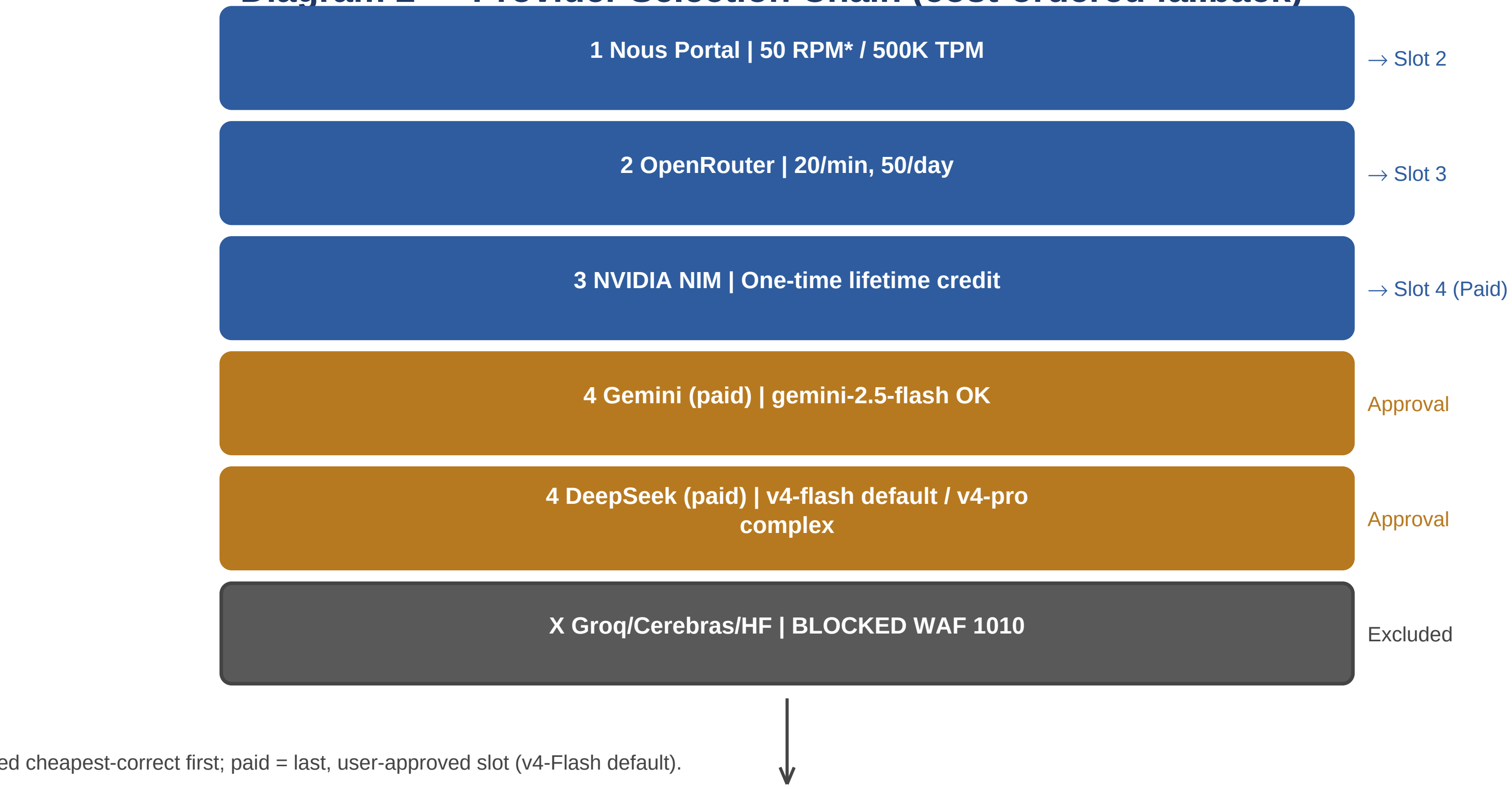
High-contrast edition. Diagram 1–5 below; Diagrams 6–7 (Interruption + 22-step loop) follow.

Diagram 1 — System Topology (layers & flow)



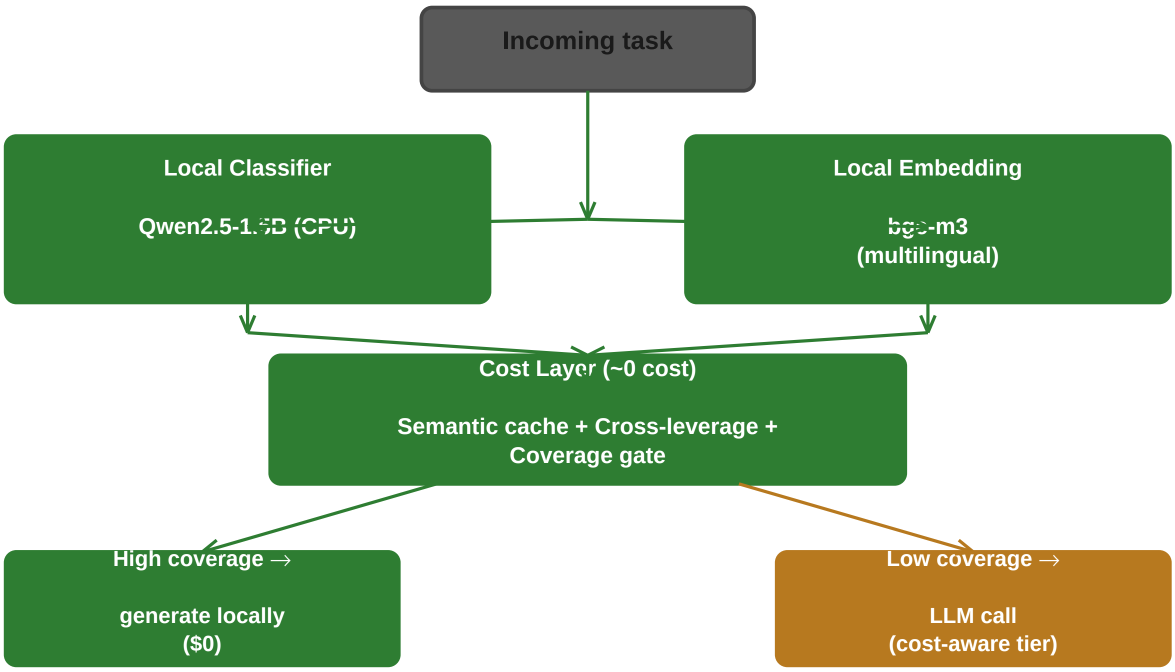
Every task flows downward through LOCAL gate + cost layer before any paid model. You approve paid escalation.

Diagram 2 — Provider Selection Chain (cost-ordered fallback)



Priority: Nous → OpenRouter → NVIDIA NIM (free) → Paid (approval). Groq/Cerebras/HF excluded (WAF-blocked).

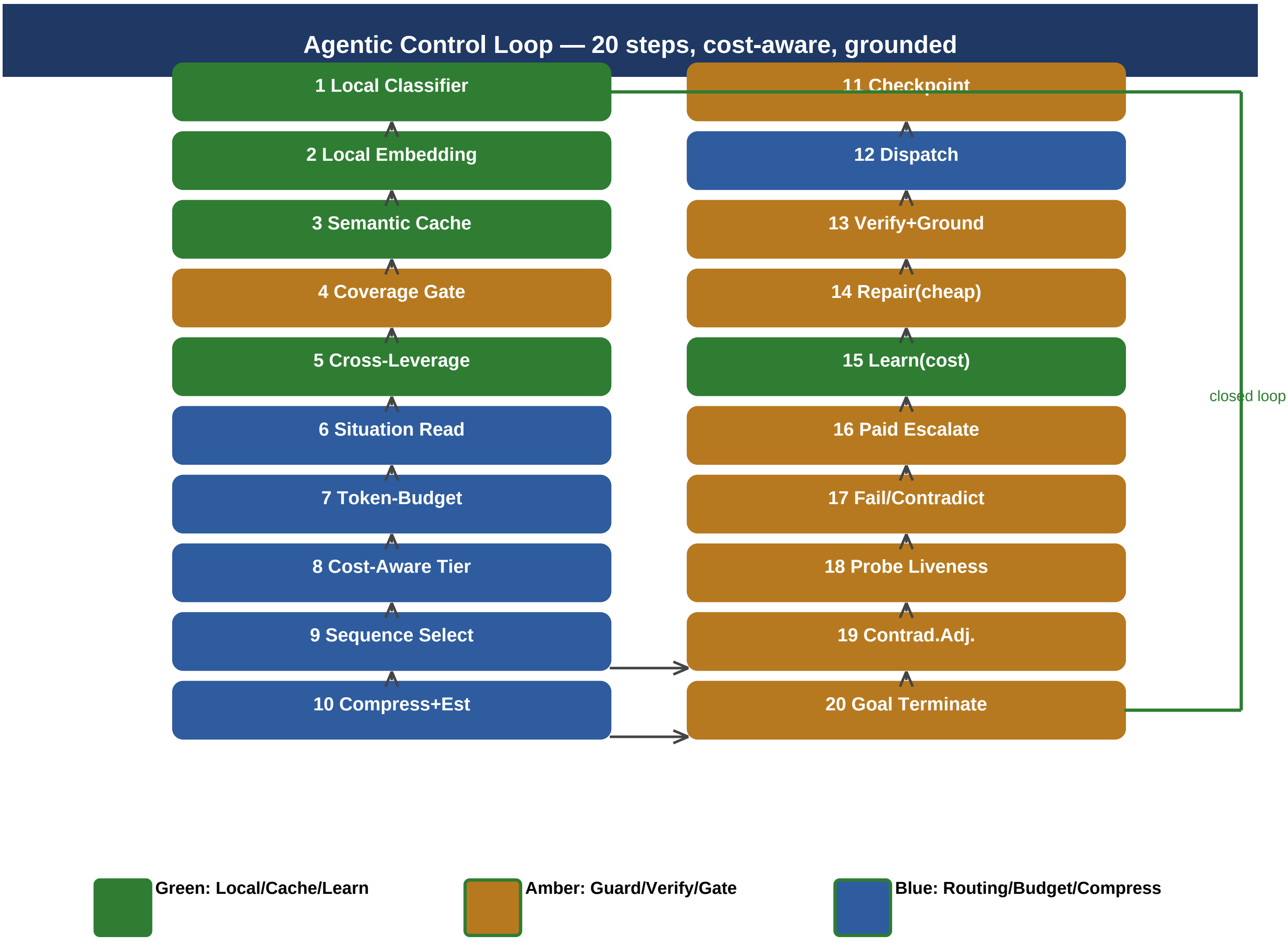
Diagram 3 — Local-First Gate (no API call until needed)



No API call until the cost-aware tier is chosen.

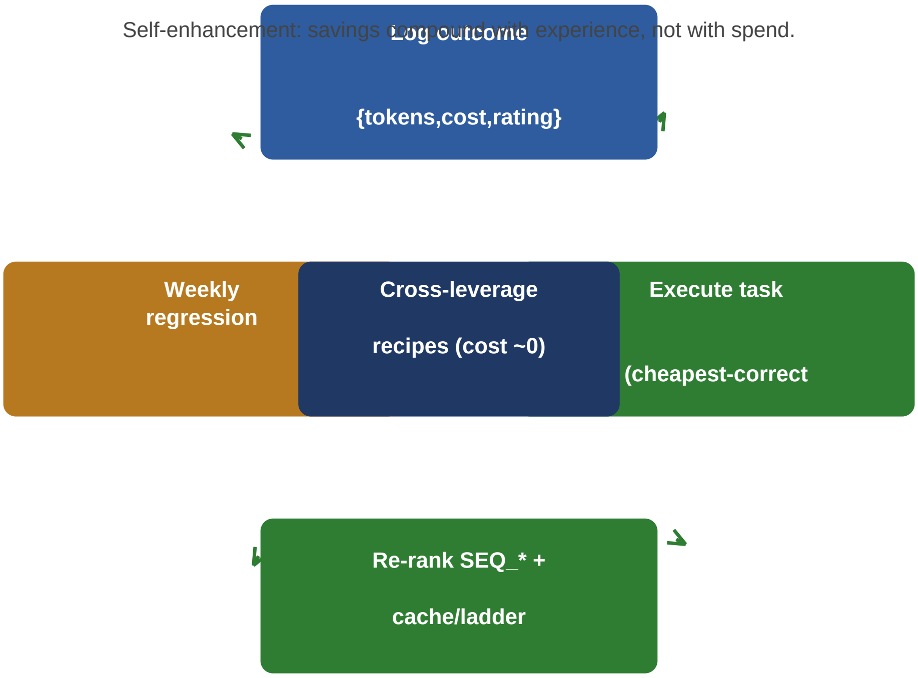
Classification + embedding run offline; cache/cross-leverage/coverage may resolve the task at $0 before any LLM call.

Diagram 4 — Agentic Control Loop (20 steps)



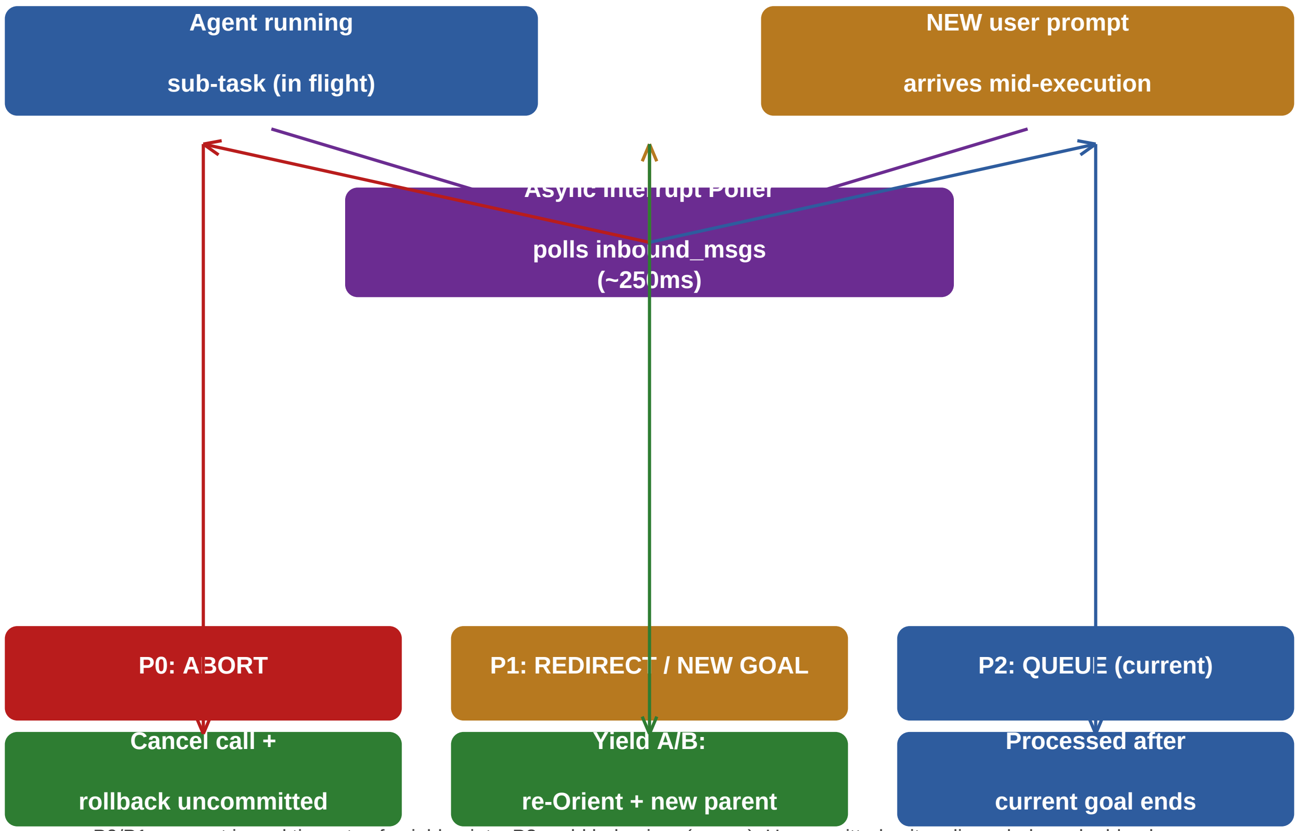
Steps 3 & 5 are ~0-cost paths; step 8 picks cheapest-correct tier; step 15 re-ranks by cost/correctness.

Diagram 5 — Cost & Learning Feedback (self-enhancement engine)



Execute → Log → Regression(cost/correctness) → Re-rank → repeat. Cross-leverage recipes promoted above LLM calls, so cost falls as experience grows.

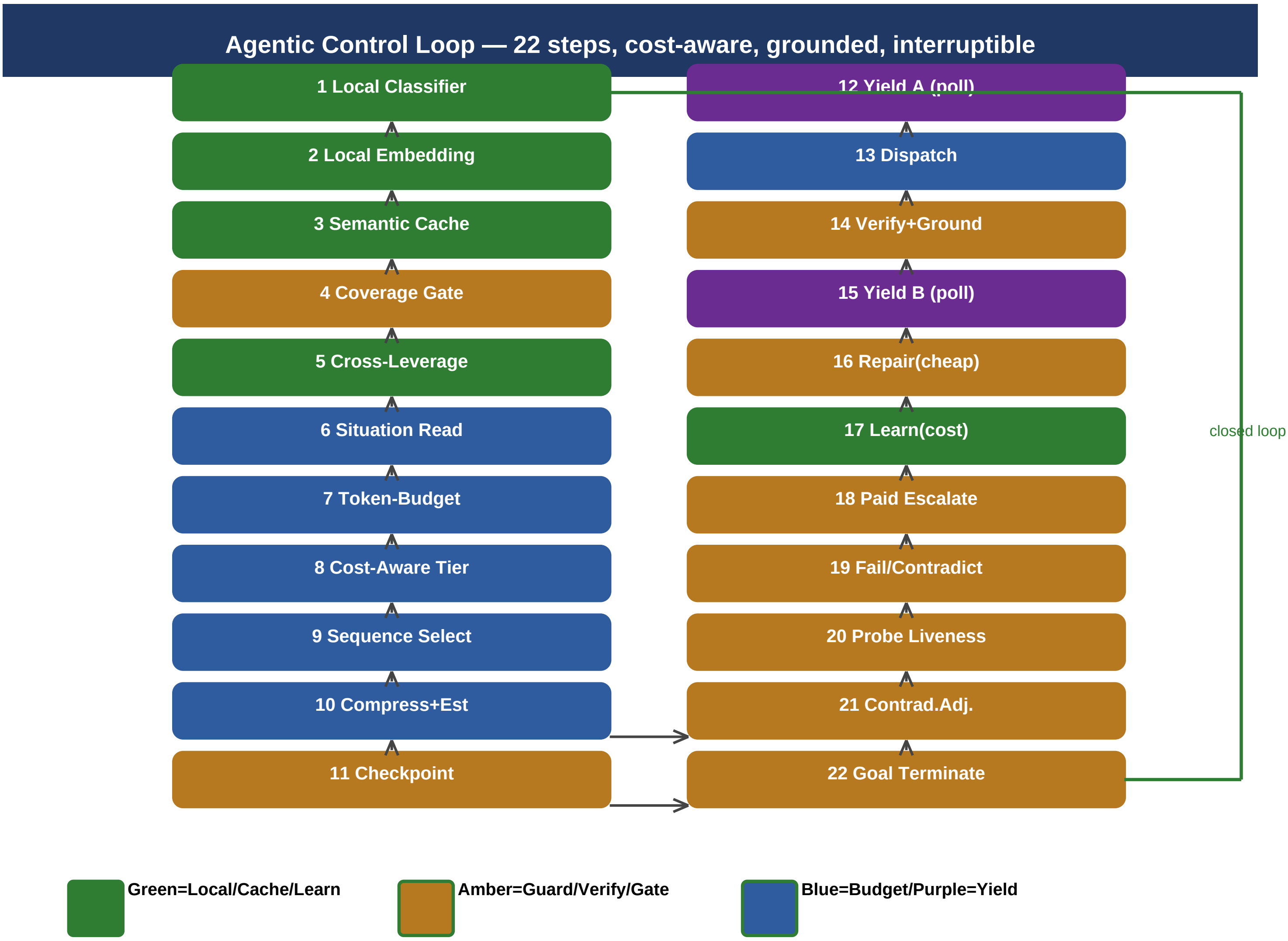
Diagram 6 — Interruption & Real-Time Maneuver



P0/P1 preempt in real time at safe yield points; P2 = old behaviour (queue). Uncommitted writes discarded; no double-charge.

A new prompt arriving mid-execution feeds an async Interrupt Poller. It splits into P0 abort / P1 re-orient / P2 queue. P0/P1 preempt at safe yield points; uncommitted writes discarded; no double-charge.

Diagram 7 — Agentic Control Loop (22 steps, interruptible)



Steps 12 (Yield A) and 15 (Yield B) are interrupt yield points polled by the async Interrupt Poller. Step 22 returns to step 1 (closed loop).